Customer Support Analytics

Analyzing customer support tickets to identify trends and business impact.

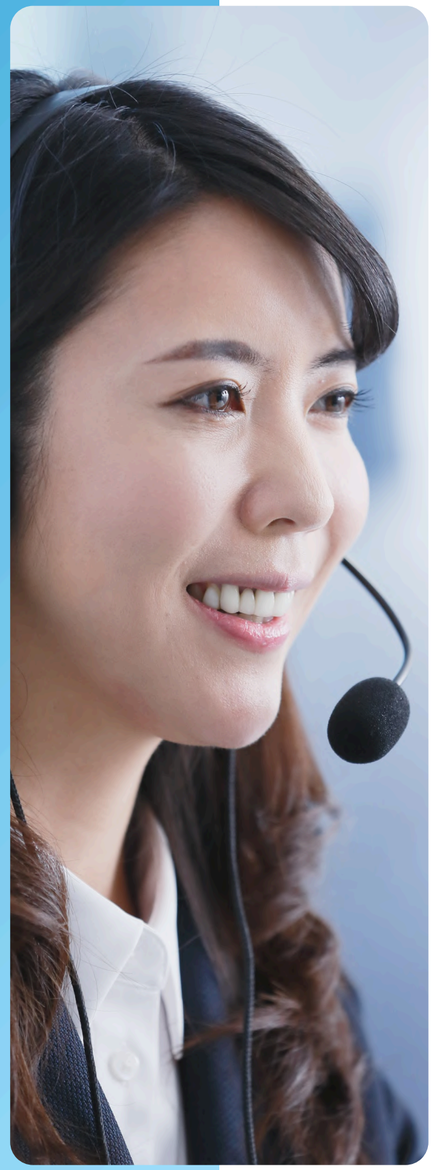
Excel | SQL | Power BI



Presented by:

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Problem Statement

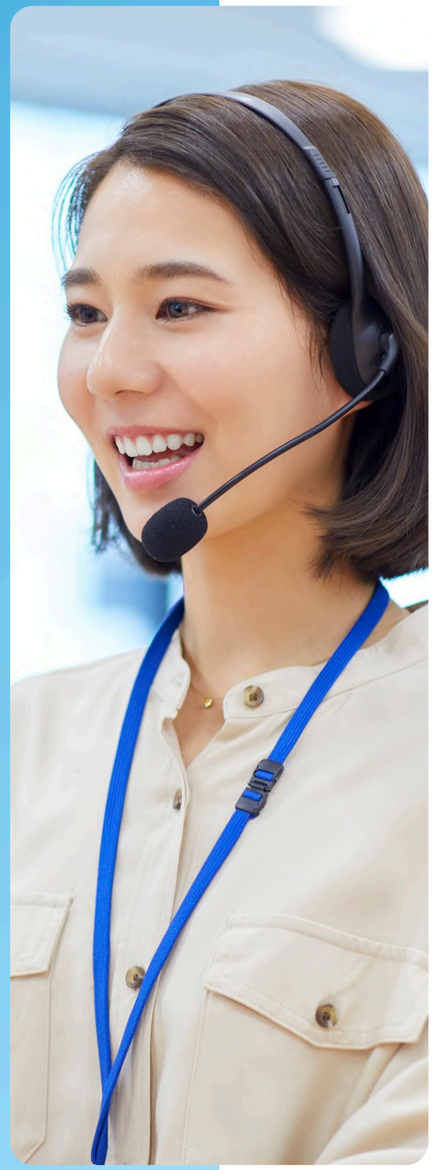


- Analyze customer support data to understand ticket trends, assess support performance and customer satisfaction, identify recurring product issues, and evaluate refunds.
- Use these insights to identify operational improvement opportunities and support data-driven business decisions.

Dataset Overview

- Source: Kaggle — E-Commerce Customer Support and Returns Dataset
- Period: August 2024 – July 2026
- Records: Approximately 25,000 support tickets
- Data type: Synthetic dataset

Key Fields



- **Ticket Details:** Ticket ID, creation date, channel, priority, resolution status
- **Product & Issue:** Product category, product name, issue category
- **Support Performance:** First response time, resolution time, CSAT score
- **Financial & Geographic:** Refund amount, region

Tools Used

- **Excel:** Data profiling, cleaning, and validation
- **SQL:** Data transformation, data modelling, and analysis
- **Power BI:** Data modelling, DAX measures, interactive dashboards, and visualization


2011
2012
2013
2014
300
225
150
Product 1
Product 2
Product 3
Product 4
0
75
150
225
300
19%
19%
25%

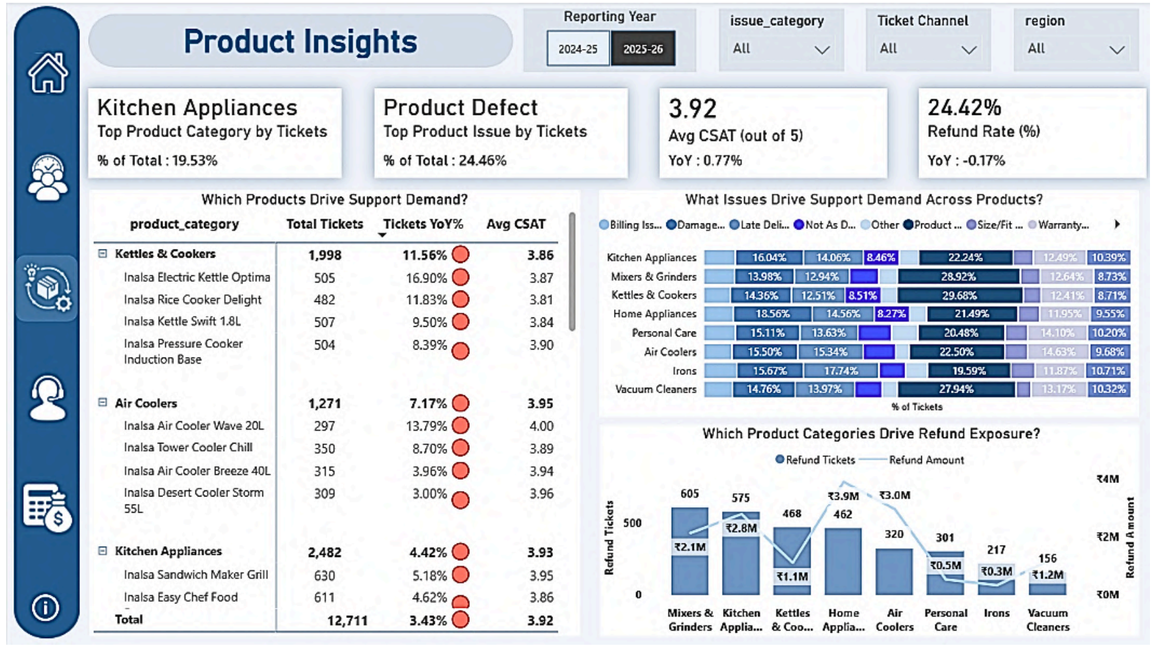
Dashboard

Dashboard Focus



Executive Overview :

Overview · Trends · Performance · Impact



Product Insights:

Product Trends · Issue Breakdown · Product Performance · Refund Drivers

Dashboard Focus

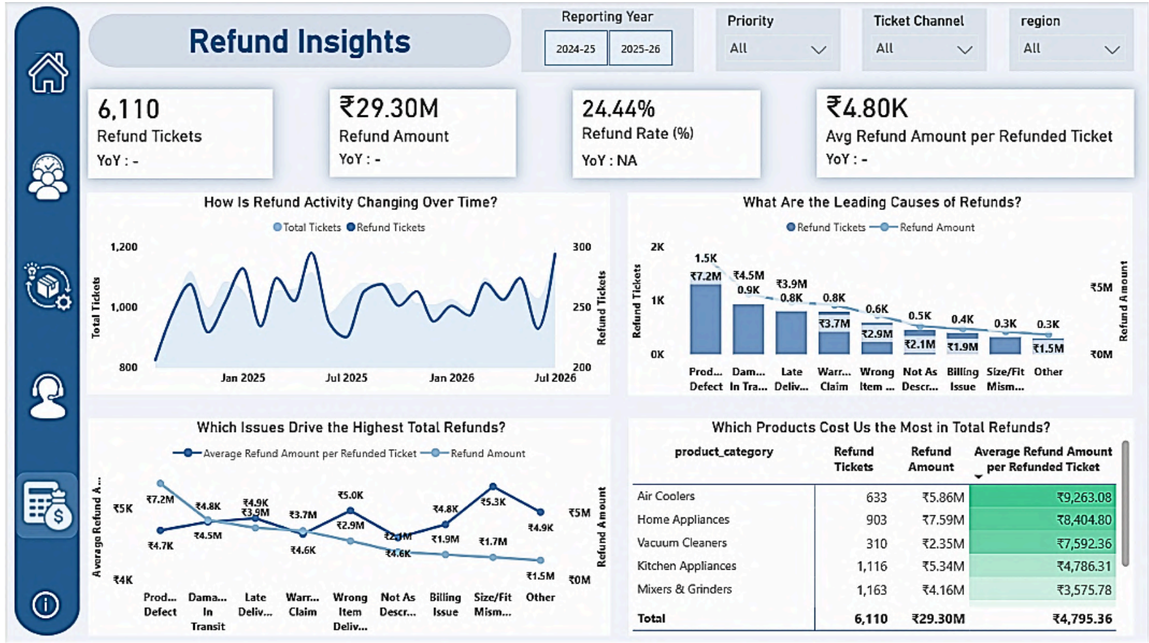


Support Performance:

Efficiency · Resolution · Satisfaction · Agent Performance

Refund Insights:

Refund Overview · Cost Analysis · Satisfaction · Financial Impact





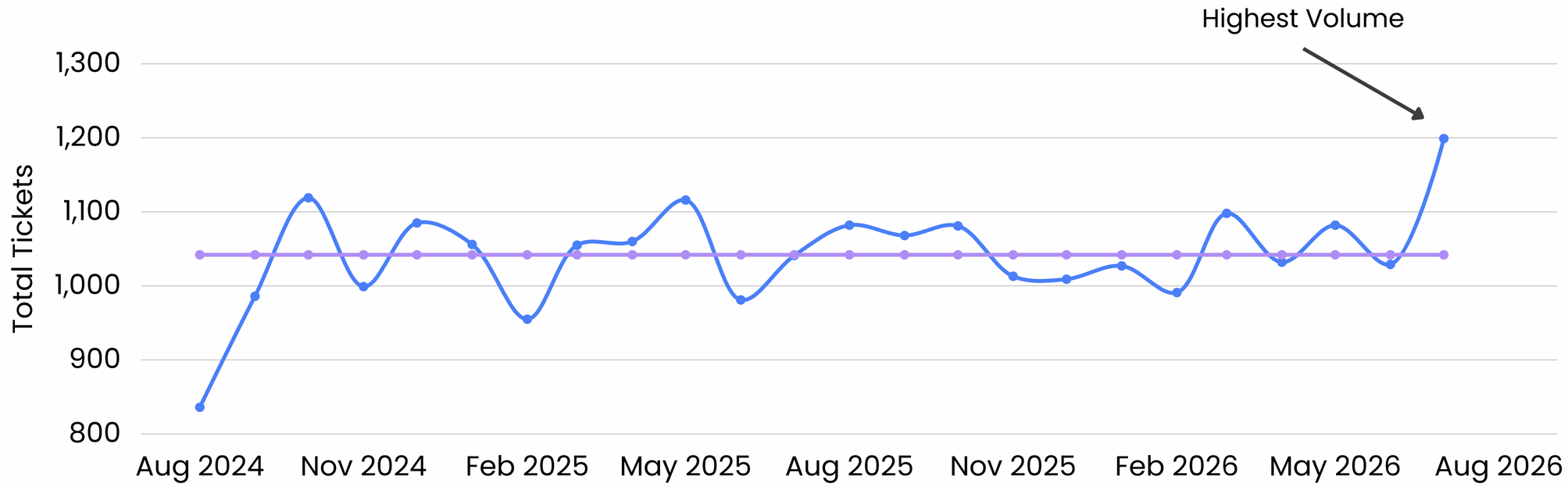
Insights

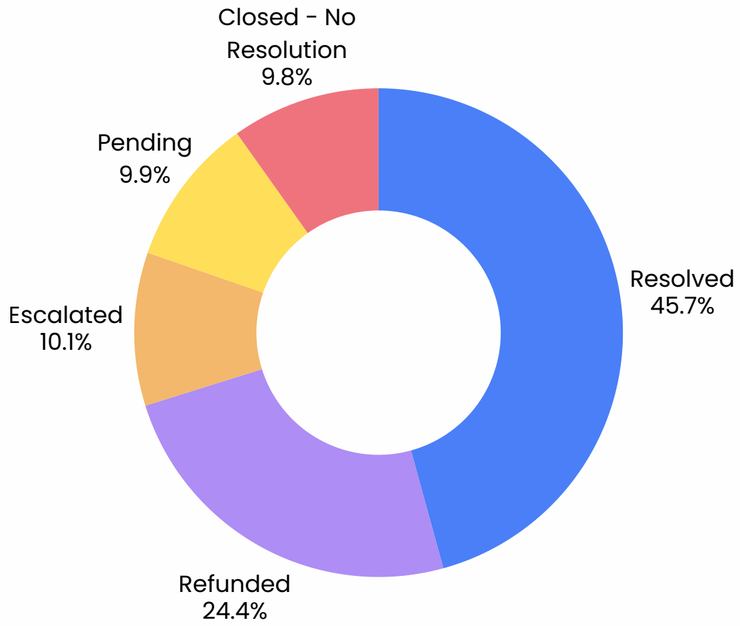
Support demand increased modestly, with July reaching a new peak

Overall ticket volume grew by **3.43% year over year**. While monthly demand fluctuated throughout both years, July 2026 recorded the highest volume at 1,199 tickets, rising sharply from June— up 16.5% from June.

Some monthly patterns repeat across both years

Ticket volumes declined in November, February and June, while October, March, May and July saw increases in both years.



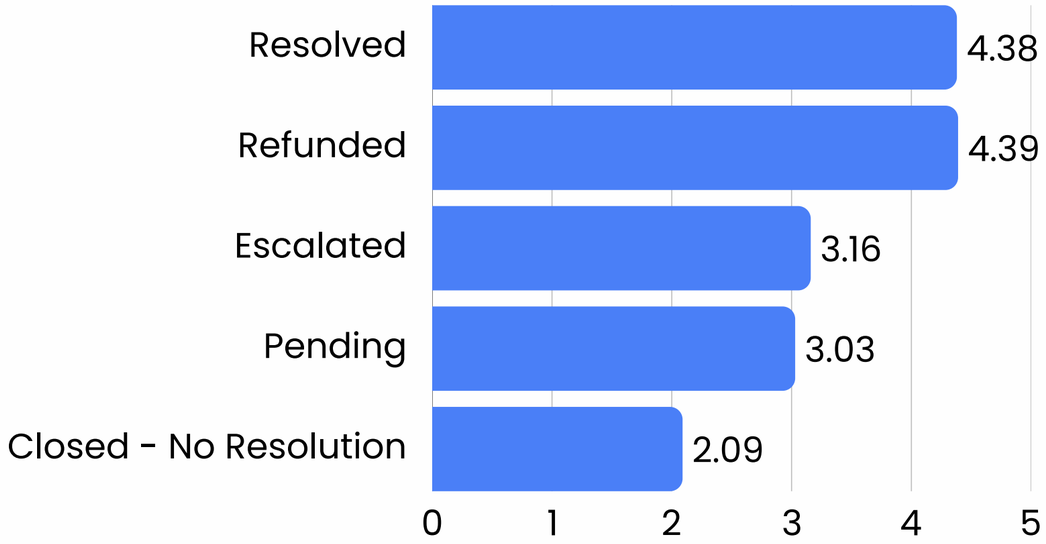


Support resolution showed modest improvement:

Support resolution improved modestly year over year, increasing by **2.32%**. However, only **45.73%** of tickets were resolved, indicating that resolution remains a key area for improvement.

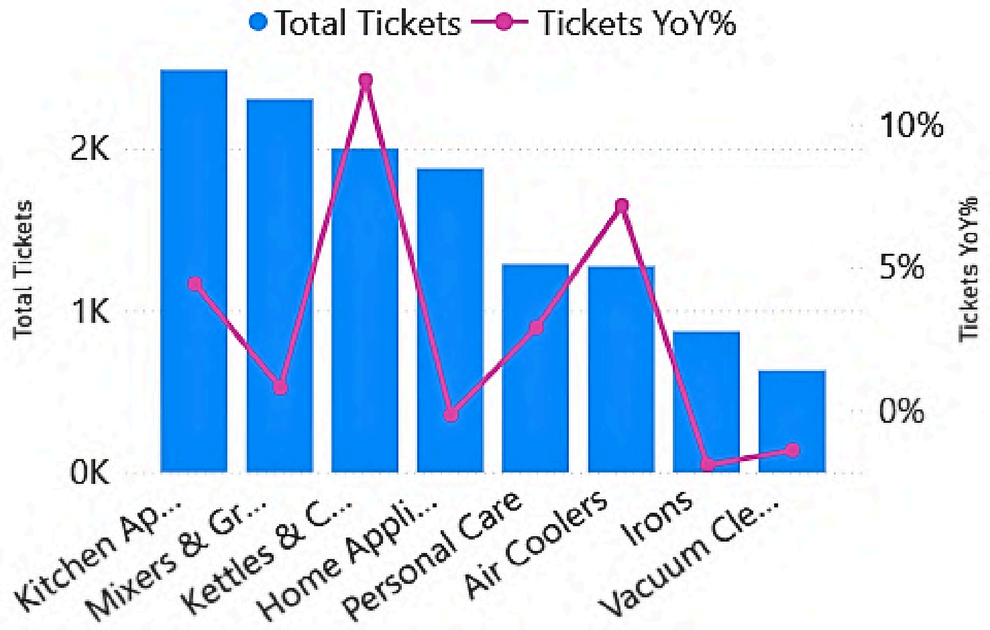
Customer satisfaction is closely linked to ticket outcomes.

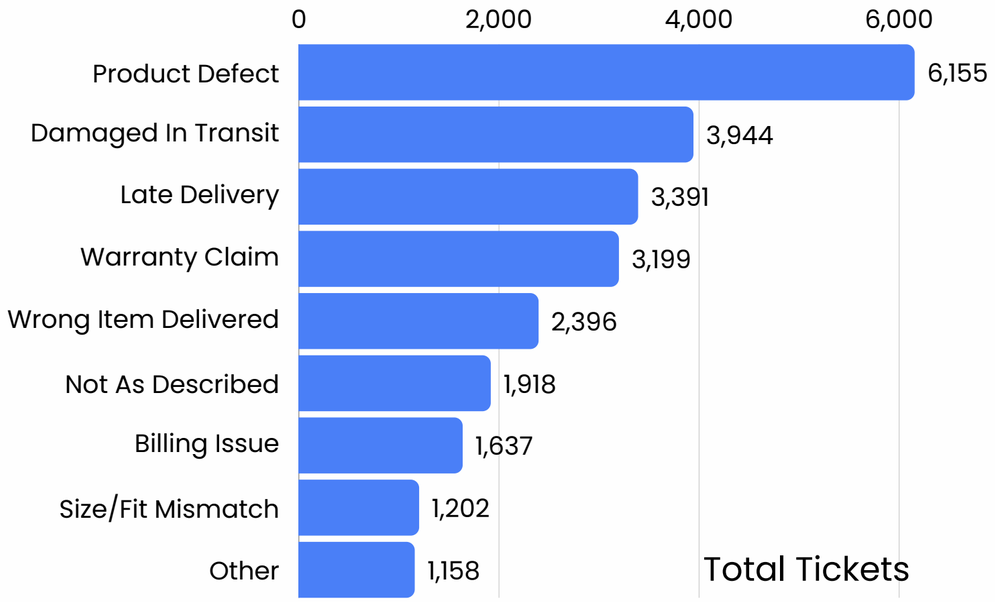
- Resolved tickets recorded an average CSAT of **4.38**, compared with other unresolved cases
- This highlights the importance of resolving customer issues successfully, as unresolved or pending outcomes are associated with lower customer satisfaction.



Support demand is concentrated around specific products

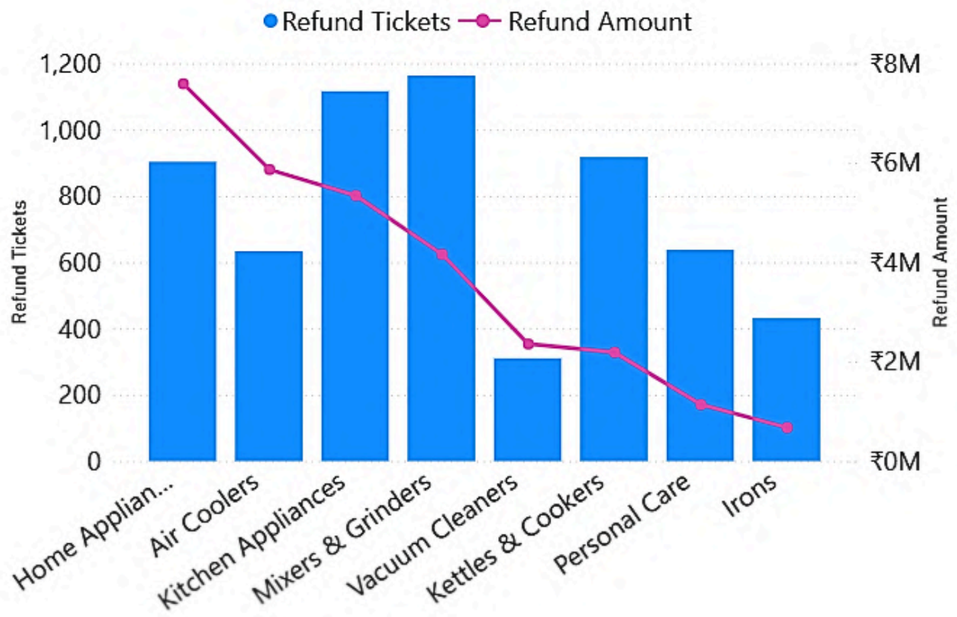
- Kettles & Cookers grew 11.56%, Air Coolers 7.17%, and Kitchen Appliances 4.42% YoY.
- **Kettles & Cookers** is the main emerging concern with rising demand and lowest satisfaction CSAT - 3.86

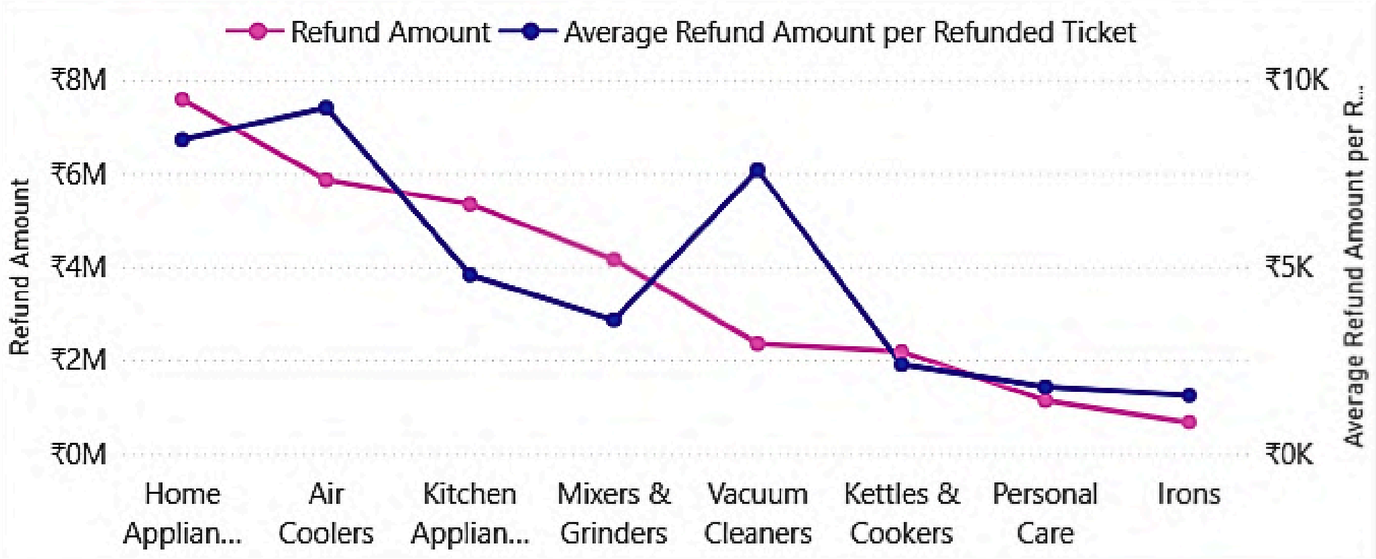




Support demand is concentrated around specific issues

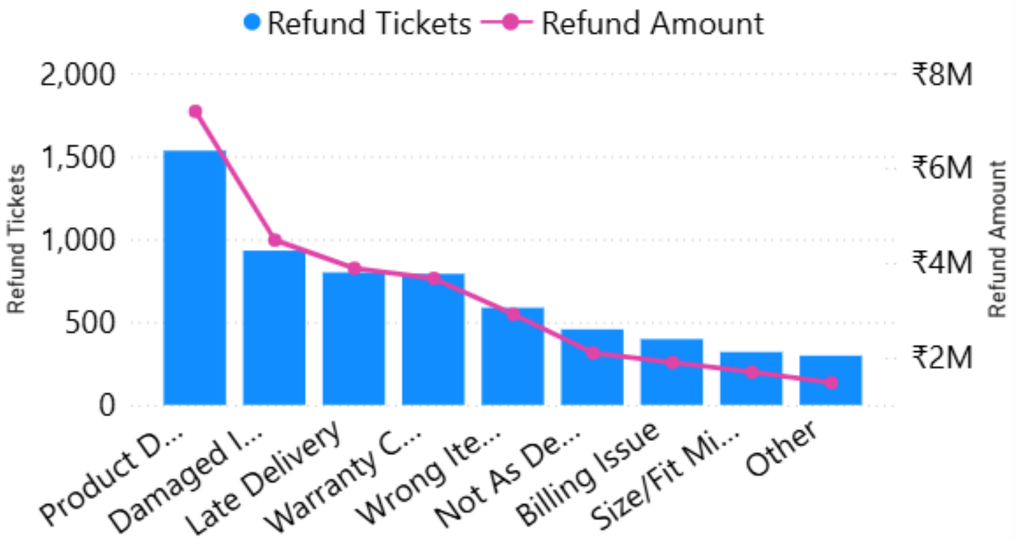
- **Product Defect** accounted for 6,155 tickets, representing 28–30% of tickets in major categories like Kettles & Cookers and Mixers & Grinders.
- Product Defect, Damaged in Transit, and Late Delivery contributed to both support workload and refund activity.

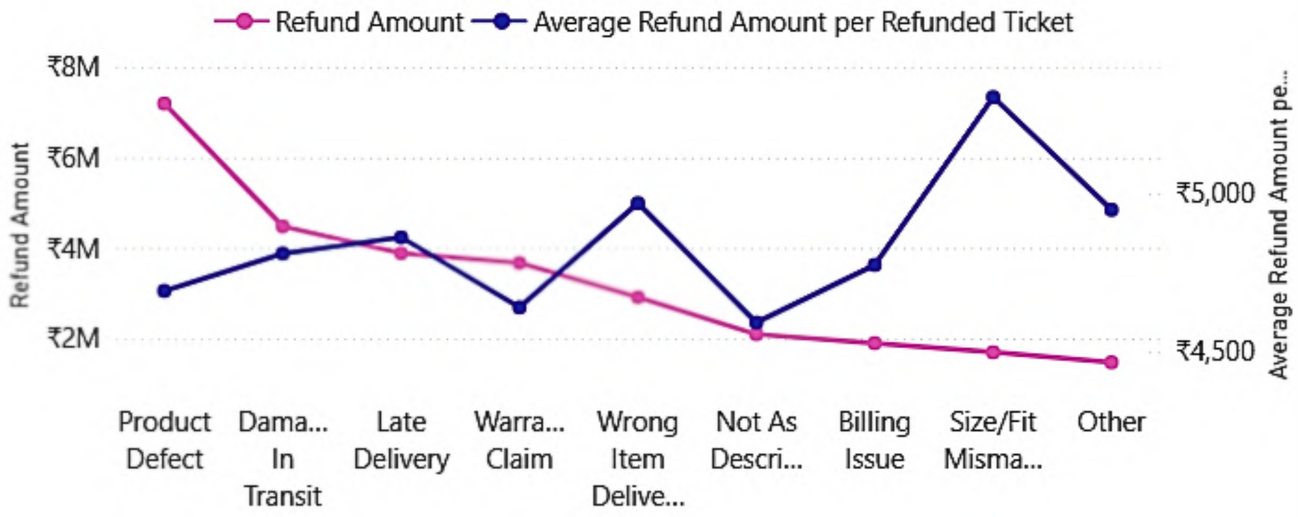




Refund exposure varies by both frequency and financial value

Mixers & Grinders and Kitchen Appliances had higher refund volumes but lower total and average refund values, while Home Appliances, Air Coolers and Vacuum Cleaners had fewer tickets but higher total and average refunds (**₹7.5K–₹9.5K**), meaning each refunded ticket costs more in these categories, increasing the company's refund expenses even with fewer tickets.





Support workload and refund costs vary by issue type

Product Defect generated the most support tickets (**6,155**) and had a 24.92% refund rate. In contrast, Size/Fit had the highest refund rate (26.54%) and average refund per ticket despite fewer tickets, highlighting Product Defect's greater workload and Size/Fit's higher refund cost per case.



Recommendations

1

Review High-Value Refund Categories

Home Appliances and Air Coolers recorded ₹7.59M and ₹5.86M in refunds, respectively. Investigate root causes and prevent avoidable refunds to reduce financial exposure.

2

Optimize Support Capacity Planning

Ticket volumes rose in October, March, May, and July. Use monthly demand patterns to plan staffing and allocate workloads proactively during peak periods.

3

Improve Ticket Resolution

Unresolved tickets like Pending tickets have lower CSAT than resolved tickets. Identify resolution barriers and address them through process improvements and targeted training.

4

Address Recurring Product Problems

Product Defects and other issues drive support demand across multiple categories. Work with Product Development and Quality teams to identify root causes and improve product quality.

5

Reduce Delivery and Fulfilment Issues

Product Defect, Damaged in Transit, and Late Delivery account for more than half of escalated cases, Work with Logistics and Supply Chain teams to address recurring causes.

Thank You for Your Attention

Explore More -

Dataset here -

